Tut- 4 (PERMUTATION) ①

① ⓐ make places for girls : 7 places Next to 6 Boys. Thus $^7P_6 \times 6!$ → Assuming 6 B.

ⓑ ~~$\lfloor 6$~~ $\lfloor 12-6+1 \; \lfloor 6 = \lfloor 7 \; \lfloor 6$

ⓒ This is the case where Boys and Girls alternate

$B_1 \; G_1 \; B_2 \; G_2 \; B_3 \; G_3 \; B_4 \; G_4 \; B_5 \; G_5 \; B_6 \; G_6$

$6! \; 6! \times 2$ → Either Boy / Girl can be the left most person.

ⓓ $\lfloor 12-6+1-6+1 \; \lfloor 6 \lfloor 6 = \lfloor 2 \lfloor 6 \lfloor 6$

② ⓐ $\dfrac{3 \times 4 \times 3 \times 2 \times 1}{\lfloor 2} = \dfrac{3 \times \lfloor 4}{\lfloor 2}$

ⓑ $\underset{\text{FIRST}}{3} \times \underset{2!}{\times 3! \times} \underset{\text{LAST}}{2} = \dfrac{3 \times 2 \times 3!}{2!}$

ⓒ Arrange D & T . $\underset{1}{\bigcirc}$ D $\underset{2}{\bigcirc}$ T $\underset{3}{\bigcirc}$

$\dfrac{^3P_3 \times \lfloor 2}{2!}$

③ We have to select 5 digit out of 6 digit but the sum should be multiple of 3.

As condition Number is divisible by 6 = (2 and 3)

Two Case: ① 0, 1, 2, 4, 5 leave (3)

sum = 12

ⓐ ends with 0 $\underset{1}{4} \times 3 \times 2 \times 1 \times 1 = 4! = 24$

ⓑ ends with (2/4) $\underset{3}{3} \times 3 \times 2 \times 1 \times \underset{24}{2} = \cancel{4 \times 2}$ 36

(II) leave 0 ; 1, 2, 3, 4, 5 sum = 15

$4 \times 3 \times 2 \times 1 \times \underset{24}{2} = 48$ → Total 108

(4) (a) $\underset{0}{\underline{9}} \times \underset{10}{\underline{\ }} \times \underset{10}{\underline{\ }} \times \underset{2,3,5,7}{\underline{4}} \Rightarrow 3600$

is Not allowed

(b) $\underset{\times\ 0\ \times}{\underline{8}} \times \underline{8} \times \underline{7} \times \underset{(2,3,5,7)\ \times}{\underline{4}} = 1792$

(5) (6)

3,4	2,3,4	2,3,4	2,3,4
D_1	D_2	D_3	D_4

Total Number $\Rightarrow 2 \times 3 \times 3 \times 3 = 54$

$\underline{\underline{>3000}}$

Thus 3,4 can be FIRST Digit

But only $2 \to 2$'s Thus 3222, 4222] can't be made

only $3 \to 3$'s Thus 3333 can't be made

Thus only $54 - 3 = 51$

(6) (5) $\underline{(2,3,4)}$ Arrangement $\underline{\ }\ \underline{\ }\ \underline{1}$ last digit is 1

Then 6 possibilities as 2,3,4 can be arranged 3! ways.

Similar way if last digit is 2

Then 6 possibilities as (1,3,4) can be arranged $6 = 3!$ ways.

Similar for 3,4

Thus sum of 1st digit is $6(1+2+3+4)$

Same for 10's digit $6(1+2+3+4)$

Thus sum $(1111) \times 6 \times (1+2+3+4)$

(Sum of digits)

$= (1111) \times 6 \times 10$.

Arrangement $(n-1)$ digits, $\Rightarrow 3! = 6$

Two cases:

(A) Ends with zero $\underline{9} \times \underline{8} \times \underline{7} \times \underline{1}$ (0)

(B) Ends with Non zero $\underline{8} \times \underline{8} \times \underline{7} \times \underline{4}$

0, x (x x) 2, 4, 6, 8 (x)

$\Rightarrow$ 2296

(8) Maximum 4 digits Number possible.

7 7 (0-9) (0-9) Except 7

1 way 1 way 9 × 9

Exactly 2 7's Thus:

$^4C_2$ $^4C_2 = 6$ places possible:

$81 \times 6 \Rightarrow 486$ ways

$\underline{9} \times \underline{7} \; \underline{7} \times \underline{9} = 9 \times 9$

(0-9) (0-9)

Except 7: Here if 1st digit is zero All 3 digit Number are Included.

$\underline{9} \times \underline{1} \times \underline{9} \times \underline{1} = 9 \times 9$

(0-9) 7 (0-9) 7

$\underline{1} \times \underline{9} \times \underline{1} \times \underline{9} = 9 \times 9$

7 (0-9) 7 (0-9)

$\underline{1} \times \underline{9} \times \underline{1} \times \underline{9} = 9 \times 9$

7 (0-9) (0-9) 7

$\underline{9} \times \underline{9} \times \underline{1} \times \underline{1} = 9 \times 9$

(0-9) (0-9) 7 7

Total

$9 \times 9 \times 6$

$= 486$

if Both 0's we will get 2 digit Number

if 1st digit is zero, we will get 3 digits Number

9 EARTH

Case 1 $\underset{A,H,E}{3} \times 4 \times 3 \times 2 \times 1$ Before ~~RA~~ RTHAE

1st ~~Digit~~ letter is before R

$\Rightarrow$ 72

$\underline{R}$ $\underset{A,H,E}{3} \times 3 \times 2 \times 1 = 18$

Case 2: 2nd letter is before T

Case 3: 3RD letter is before H

$\underline{R}$ $\underline{T}$ $\underset{A,E}{2} \times 2 \times 1 = 4$

Case 4: RTHAE $\Rightarrow$ 1

as No letter come befor A

Thus Rank of RTHAE $\Rightarrow 72+18+4+1$

$\Rightarrow$ 95

10

1	2	3	4	5	6

3 Vowels 3 places $\rightarrow$ 1, 3, 5 $\Rightarrow$ $\lfloor 3$.

$\Rightarrow$ Remaining 3 place – $\lfloor 3$

$= \lfloor 3 \times \lfloor 3$.